

CHUNGBUK NATIONAL UNIVERSITY · SOFTWARE ENGINEERING · GROUP 4

AI Language Village

An immersive 3D virtual environment for **spoken English acquisition** powered by autonomous Generative AI NPCs and real-time speech evaluation.

React 19 · Three.js + R3F · Vite 8 · Groq Cloud LLM · Web Speech API · Firebase Auth

01 · EXECUTIVE SUMMARY

Bridging the Gap to Spoken Fluency



Spatial Immersion

12 fully modeled 3D facilities across Commercial, Civic, Healthcare, and Campus districts in a living village.



12 AI Personas

Autonomous LLM agents with distinct backstories, vocabulary domains, and dynamic unscripted dialogue.



Zero-Lag Voice Loop

In-browser Web Speech STT, sub-600ms Groq LLM inference, and synchronized 3D avatar speech animation.



Diagnostic Examiner

Intelligent CEFR assessment filtering microphone noise and delivering native-language grammatical feedback.

02 · THE PROBLEM

The "Silent English" Paradox



High Speaking Anxiety

Over **70% of language learners** experience acute psychological friction when speaking with humans, fearing peer judgment and social embarrassment.



Passive, Gamified Apps

Traditional apps (Duolingo, Quizlet) rely on multiple-choice taps and isolated vocabulary cards—completely lacking situational context and conversational spontaneity.



Prohibitive Tutoring Costs

1-on-1 private tutoring with native speakers costs **$30 to $60 per hour**, making continuous, daily conversational immersion financially unattainable.

03 · THE SOLUTION

Spatial Game Engine + Generative AI



Zero-Anxiety Practice Sandbox

Learners converse with non-judgmental AI NPCs in an infinite, patient environment. Mistakes become constructive learning opportunities with zero social cost.



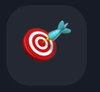
Spatial Memory Retention

Cognitive science proves vocabulary retention increases by over **40%** when anchored to spatial 3D environments (e.g., ordering coffee inside an authentic cafe).



Authentic Unscripted Discourse

No rigid decision trees. Groq-powered LLMs adapt naturally to any utterance, accent variation, or unexpected conversational tangent in real time.



Task-Based Language Teaching (TBLT)

Every facility provides authentic mission goals: opening bank accounts, reporting lost items, consulting doctors, and practicing job interviews.

04 · CORE GAMEPLAY LOOP

The 5-Step Learning Flow

01



Explore

W A S D + Mouse

MiniMap Radar

→

02



Enter

Approach door

Press E

→

03



Briefing

Contextual Mission

Goal & Task Card

→

04



Dialogue

Press F to talk

Voice STT & TTS

→

05



Evaluation

Exit with E

Score, Errors & XP

05 · VIRTUAL WORLD ARCHITECTURE

12 Interactive Facilities Across 4 Districts



Village Shop
NPC: Jack (Shopkeeper)
Groceries, prices, count nouns & quantities.



Language Cafe
NPC: Mia (Barista)
Beverages, pastry orders, polite requests.



Village Bank
NPC: David (Banker)
Accounts, wire transfers, debit cards & fees.



Police Station
NPC: Officer Alex
Lost property reports, emergency advice.



Village Hospital
NPC: Sarah (Receptionist)
Doctor appointments, triage & wait times.



Village Hotel
NPC: Emma (Front Desk)
Room reservations, check-in, amenities.



Language School
NPC: Teacher Olivia
English proficiency classes & curriculum.



Study Center
NPC: Tutor Daniel
Study strategies, habits & vocabulary tips.



Career Center
NPC: James (Counselor)
Job interviews, resumes & vocational training.



Village Library
NPC: Emily (Librarian)
Book borrowing, catalog searches & cards.



Community Center
NPC: Mina (Civil Staff)
Residence registration & community programs.



Activity Center
NPC: Hana / Yuri
Social clubs, workshops & cultural events.

06 · AUTONOMOUS AI AGENTS

Persona-Driven Conversational Intelligence

🎭

Specialized Domain Vocabularies

Each agent is bound to a tailored system prompt specifying domain jargon (financial terms for David, medical triage for Sarah, hospitality etiquette for Emma).

💬

Conversational Pacing & Length

Prompts strictly restrict replies to **1–2 natural spoken sentences** without quotation marks or markdown, ensuring realistic verbal conversation rhythms.

🔄

Rolling Memory Buffer

Agents maintain a rolling 4-turn message memory buffer, remembering preceding context while keeping token overhead and latency strictly minimized.

```
// Dynamic Persona Prompt Injection (App.jsx)
const systemPrompt =
  npc === "Bank Employee" ? `You are David, a polite bank officer at Village Bank. Help with accounts, wire transfers, cards. 1-2 short sentences.`
  : npc === "Career Counselor James" ? `You are James, a friendly career counselor. Help with job searching, resumes, and interview prep.`
  : `You are Mia, a friendly cafe barista in a 3D village. Help the user order drinks and pastries naturally.`;
```

07 · PEDAGOGICAL METHODOLOGY

48+ Situational Missions Mapped to CEFR Levels

A1–A2 · Elementary

Direct Transactions & Routines:

- Shop: Buy 2 bottles of water and 1 apple.
- Cafe: Order 1 Americano & 1 croissant.
- Library: Ask where beginner books are.

B1 · Intermediate

Inquiries & Administrative Scenarios:

- Hotel: Reserve room for 2 nights + breakfast.
- Bank: Open account & ask required documents.
- Police: Report a lost phone and file report.

B2–C1 · Advanced

Professional & Nuanced Discourse:

- Career Center: Mock job interview coaching.
- Hospital: Describe symptoms & doctor triage.
- Study Center: Formulate a study schedule.

08 · TECHNICAL ARCHITECTURE

The Real-Time Voice AI Pipeline

VOICE INPUT



Speech-to-Text

Web Speech API

`webkitSpeechRecognition`

FAST INFERENCE



Groq Cloud LLM

`openai/gpt-oss-120b`

~480ms response time

SPEECH & AVATAR



TTS & 3D Sync

`speechSynthesis` +

Avatar mouth/head motion

ON FACILITY EXIT



Examiner AI

Diagnostic rubric

Strict JSON evaluation

Sub-600ms Round-Trip Response

Groq's Tensor Streaming Processors execute 120-billion parameter models in real time, matching human conversational cadence without awkward latency pauses.

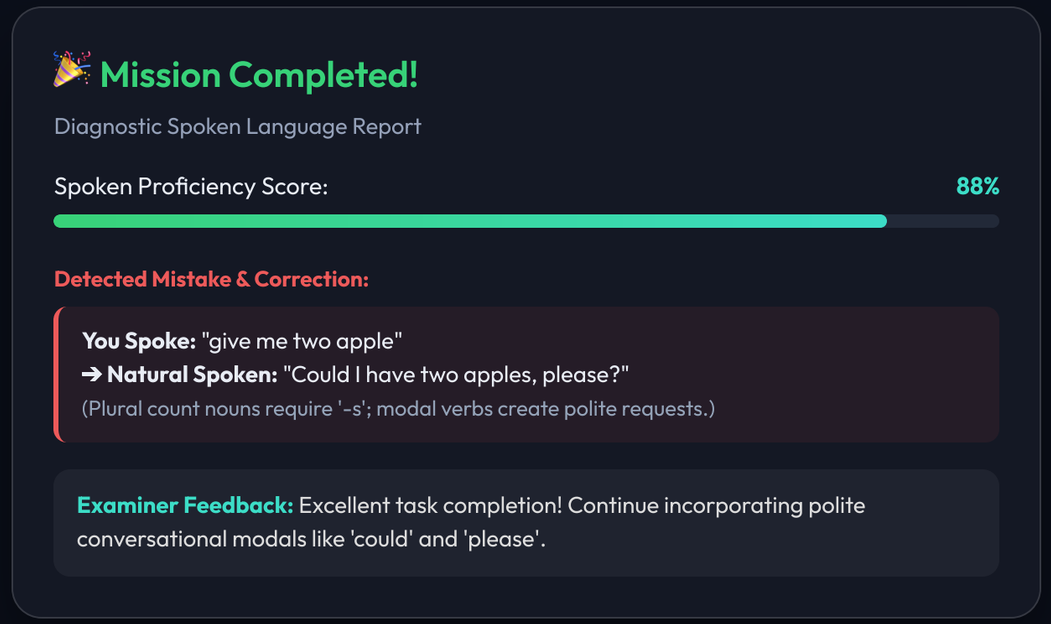
Zero Server Audio Bandwidth

By executing speech recognition and synthesis natively inside the browser, raw audio never leaves the client, eliminating audio server streaming costs.

09 · AUTOMATED ASSESSMENT

Speech-Aware Examiner AI

- **Disregards STT Artifacts:** Microphones omit capital letters and periods. The rubric explicitly ignores casing, preventing false negatives.
- **Task Goal Verification:** Checks that requested items, quantities, and questions were communicated accurately.
- **Real Spoken Grammar Diagnosis:** Catches count/mass errors ("two apple"), agreement slips ("I wants"), and polite modal forms.
- **Structured JSON Output:** Deterministic schema with score (0–100), completion status, and native explanations.



10 · LEARNER RETENTION

Gamification, Profile & Progress Tracking

Firebase Auth

Secure authentication with email/password, auto-session persistence, and cloud Firestore synchronization.

Onboarding Placement

Personalized CEFR level selection (A1 to C1), user study motivations, and 3D character avatar customization.

Dynamic XP & Levels

Calculates XP based on score accuracy (85%+ bonuses), mission complexity, and consecutive daily learning streaks.

MiniMap Radar

Real-time top-down 2D radar displaying player coordinates, orientation, 12 landmarks, and click-to-teleport.

Progress Panel Dashboard

Tracks facility completion rates, historical score averages, and maintains a chronological mistake log for spaced repetition.

`[ProgressPanelModal.jsx]`

Decoupled Multi-Tier Architecture

TIER 1 · UI OVERLAY (REACT 19)

HUD & Instructions	HUD
MiniMap Radar	MiniMap
Mission Briefing	Modal
Speech Chat Window	Dialog
Diagnostic Report	Examiner
Progress Dashboard	Progress

APP.JSX

Central Controller

Coordinates state machines, camera transitions, and AI client orchestrations

currentLocation	dialogState
activeMission	exitEvaluation
playerRef	cameraAnglesRef

TIER 2 · 3D SCENE (THREE.JS / R3F)

Environment	Roads, Trees, Lights
Traffic System	Moving Car Mesh
12 Buildings	Kenney GLB + AABB
Player Mesh	WASD + Animations
Camera Controller	GTA-style Orbit

Key Technical Challenges & Solutions



STT Transcription Quirks

Challenge: Microphones omit capitalization; standard LLMs penalize lowercase text as grammatical errors.
Solution: Custom evaluation rubric explicitly strips punctuation/casing sensitivity.



LLM Markdown Hallucination

Challenge: Cloud LLMs sometimes wrap JSON in markdown backticks, breaking `JSON.parse()`.
Solution: Multi-pass regex sanitization and schema-guaranteed fallback object generation.



Camera Collision & Lerping

Challenge: Moving between open roads and compact facility interiors caused camera clipping.
Solution: Contextual spherical coordinate lerping (theta/phi/radius) tuned per building.



Asset Weight & Frame Rates

Challenge: 15+ GLB models risked high memory footprint and WebGL frame drops.
Solution: Geometric instancing, shadow-bias tuning to prevent acne, and React Suspense loading.



Boundary Collisions (AABB)

Challenge: Complex 3D mesh collisions caused physics stutter.
Solution: Configured lightweight 2D AABB bounding boxes (`VILLAGE_BUILDINGS`) for smooth 60 FPS.



Speech-Avatar Sync

Challenge: Syncing synthetic voice with 3D character gestures.
Solution: Reactive `isTalking` hooks bound to TTS `onstart` and `onend` audio lifecycle events.

13 · QUANTITATIVE MILESTONES

The Project in Numbers

0	0	0	0
Lines of Core Logic	Modular 3D Components	Modeled Facilities	Autonomous AI NPCs

⚡ ~480ms LLM Latency

Average Groq API inference time, achieving instantaneous verbal response rates well below conversational awkwardness thresholds.

🎮 Solid 60 FPS

Consistent WebGL rendering performance on standard modern laptops without requiring dedicated discrete GPUs.

💰 $0 Audio Server Costs

Zero cloud bandwidth expenditure for audio streams by utilizing native client-side browser Web Speech APIs.

14 · FUTURE OUTLOOK

Product Development Roadmap

✅ PHASE 1 (COMPLETED)

Core Experience

- 3D Open-world Village
- 12 Facilities & AI NPCs
- Voice STT & TTS Pipeline
- Examiner & Firebase Auth

PHASE 2 (Q2 2027)

Phonetics & Co-Op

- WebAudio Phoneme Analysis
- Pronunciation Pitch Feedback
- Multiplayer Co-Op Roleplay
- Dynamic Weather & Seasons

PHASE 3 (Q3 2027)

Performance & Mobile

- Serverless Secure Proxy
- Draco 3D Asset Compression
- Mobile Touch & PWA Support
- Offline Scenario Caching

PHASE 4 (Q4 2027)

Enterprise & Languages

- Teacher Analytics Portal
- Custom Scenario Builder
- Korean / Japanese Learning
- University LMS Integration

INTERACTIVE WALKTHROUGH

Live Demonstration

Experience the village firsthand: explore the streets, order in the cafe, consult the bank, and review your diagnostic speaking score.

```
$ npm install && npm run dev
```

Thank you for your time and attention! We look forward to your questions. 🙌